# The self-halting set K is computably enumerable but not decidable

The Open Logic Project contributors

openlogic:computability/computability-theory:local-snapshot-20261013; CC BY 4.0

*Editorial scope.* Statement and complete author proof are included verbatim as the span lines 1677-1696 of the pinned original source; no text is rewritten or paraphrased.

*Allowed standard prerequisites (editorial).*

- partial computable functions and their indices cf_e
- the characteristic function of a set
- Kleene's T predicate and unbounded minimization
- enumerability as the domain of a partial computable function

**Theorem 0.1.** *The self-halting set $K = \{ e \mid \varphi_e(e)\downarrow \}$ is !!c.e. but not decidable.*

*Proof.* Suppose $K$ is decidable, i.e., its characteristic function $\chi K$ is computable. Let

$$d(e) = \begin{cases} 1 & \text{if } \chi K(e) = 0 \\ \uparrow & \text{otherwise.} \end{cases}$$

Let $k$ be the index of $d$, i.e., $d \simeq \varphi_k$. Then $d(k) \simeq \varphi_k(k)$. This contradicts the fact that $d(k)\downarrow$ iff $\varphi_k(k)\uparrow$, which follows from the definition of $d$.

$K$ is the domain of $f(x) = \mu y T(x, x, y)$ and so is !!c.e. □

*Proof.* Suppose $K$ is decidable, i.e., its characteristic function $\chi K$ is computable. Let

$$d(e) = \begin{cases} 1 & \text{if } \chi K(e) = 0 \\ \uparrow & \text{otherwise.} \end{cases}$$

Let $k$ be the index of $d$, i.e., $d \simeq \varphi_k$. Then $d(k) \simeq \varphi_k(k)$. This contradicts the fact that $d(k)\downarrow$ iff $\varphi_k(k)\uparrow$, which follows from the definition of $d$.

$K$ is the domain of $f(x) = \mu y T(x, x, y)$ and so is !!c.e. □